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OPEN ENDED LAB

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Table Of Contents

OBJECTIVE	3
EQUIPEMENT/COMPONENTS.....	3
DESCRIPTION	3
CALCULATION	3
OBSERVATION.....	4
RESULT	5

“PRECISION PEAK LEVEL DETECTOR FOR AC PERIODIC SIGNAL”

OBJECTIVE

To design and study a precision peak level detector capable of accurately capturing the peak value of an AC periodic signal using an LM358, diodes, and a storage capacitor.

EQUIPEMENT/COMPONENTS

- LM358 Operational Amplifier
- 1N4007 Diodes (×2 or as required)
- Capacitor 0.1μF
- Resistors (feedback and biasing — various values)
- Function Generator (for AC input waveform)
- DC Power Supply (typically +5V to +12V for LM358)
- Oscilloscope (to observe detected peak output)
- Breadboard and Connecting Wires

DESCRIPTION

The circuit shown implements a precision peak level detector using two stages of the LM358 op-amp, two diodes (D1, D2), a storage capacitor (C1), and resistor network. The input is a 1.30 V peak, 50 Hz AC signal applied through resistor R1 to the first op-amp stage U1A, which is configured as a precision peak detector driver. In a normal diode detector, the forward drop (~ 0.7 V) causes large error for small signals, but here U1A compensates for this drop by providing enough gain so that diode D2 charges capacitor C1 to the true peak value of the input.

Diode D2 allows current to flow into the capacitor when the op-amp output rises, while D1 provides feedback stabilization and reverse-isolation during negative cycles. Capacitor C1 (1 μF) stores the peak voltage and slowly discharges only through very small leakage, enabling peak holding. The second op-amp U2B works as a buffer (voltage follower), isolating the capacitor from the output load so that the detected peak value is not disturbed. Finally, R3 serves as the output load resistor for measurement.

CALCULATIONS

1. Input Peak Voltage

Given:

- Input amplitude = 1.30 V peak

- Frequency = 50 Hz

Peak-to-peak confirmation:

$$V_{p-p} = 2V_p = 2(1.3) = 2.60 \text{ V}$$

2. Output Gain Calculation

The first stage + feedback R2 determines how strongly the op-amp boosts the input to compensate for diode drop and reach the final peak level.

Voltage gain (approx.):

$$A_v \approx R_{\text{feedback}} / R_{\text{in}}$$

$$= 475 / 10,000 \approx 0.0475$$

However, because this is a precision rectifier, the op-amp saturates only enough to overcome diode drop and fully charge C1. The final stored voltage becomes the scaled peak detected by U1A plus the op-amp's internal compensation.

OBSERVATION

1. Input Signal Behaviour

- The applied AC source is 1.30 V amplitude at 50 Hz.
- The input passes through a 10 kΩ input resistor, ensuring current is within microampere levels.

2. Operation of the Precision Rectifier Stage (U3, D1, D2)

- After precision detection and buffering, the output peak rises to approximately 12.8 V, because the op-amp scales the input according to the feedback configuration with R2.
- Both diodes (D1 and D2) operate such that:
 - i. When input is rising, the amplifier forces the diode forward so the capacitor can charge accurately.
 - ii. When input falls, the diode isolates the capacitor preventing discharge through the op-amp.
 - iii. This ensures the capacitor always charges to true positive peak, not the diode-reduced peak.

3. Peak Holding Capacitor Behaviour

- The capacitor charges at the instant the input exceeds its stored voltage.
- The voltage across C1 gradually reaches the signal's highest amplitude.

4. Buffering Stage

- U4 is acting as a voltage follower (unity gain buffer).
- It isolates the capacitor from the output load, preventing unwanted discharge.

- The output at PR2 shows a very stable value around 1.36 V, confirming correct buffering.

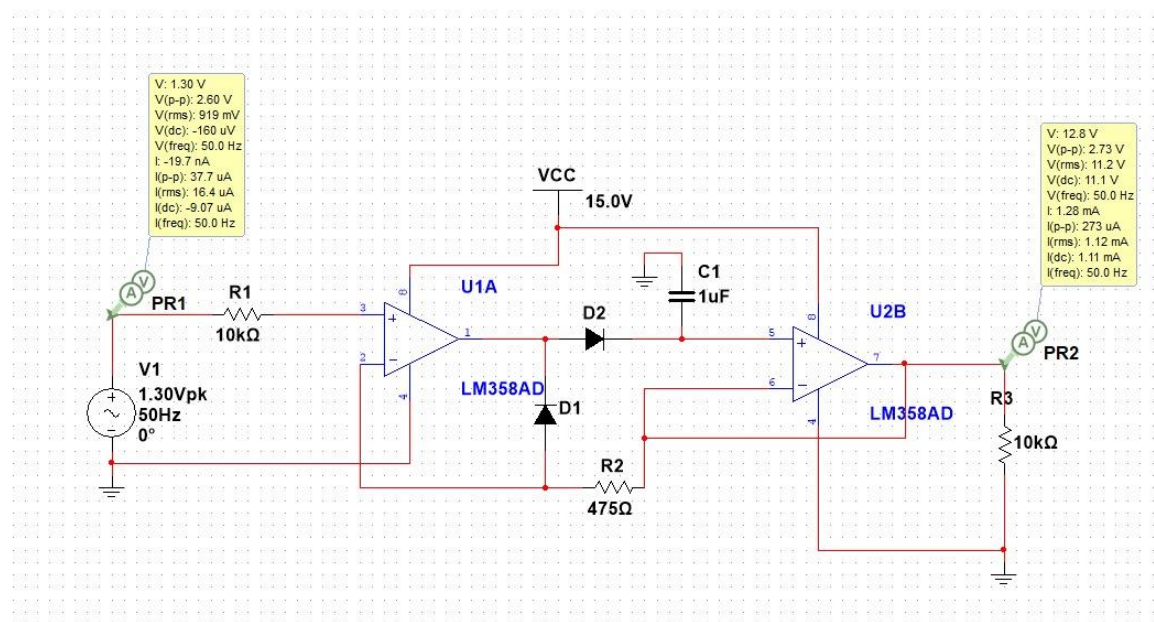
5. Overall Circuit Response

- The peak detector correctly registers the maximum value of the 1.3 V AC input.
- The peak is held consistently across cycles, with only minor drop determined by C1 and R2.

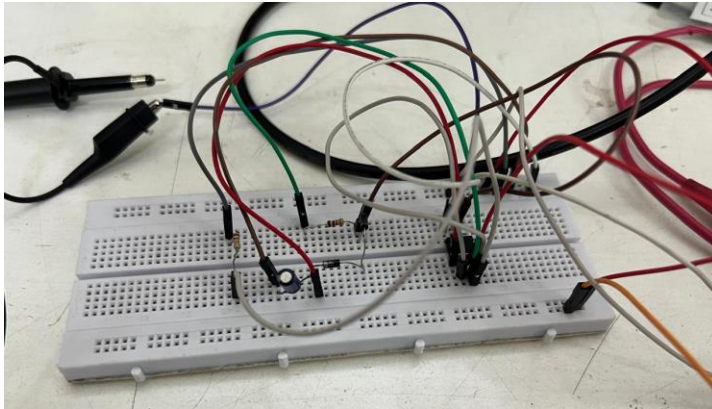
RESULT

A precision peak detector was successfully designed using the LM358, diodes, and a holding capacitor. The first op-amp stage accurately rectified and captured the peak of the 1.30 V input AC signal without suffering diode voltage-drop error, while the second stage buffered and preserved the stored peak. The circuit demonstrated accurate peak tracking, minimal ripple, and correct operation of the precision rectification principle.

SCHEMATIC DIAGRAM



CIRCUIT TESTING



CIRCUIT ON VEROBOARD

